

Author Contributions Statement

IEEE Transactions on Power Delivery

“Bayesian Three-Hypothesis SPRT for IBR-Aware Transformer Differential Protection: Waveform Asymmetry, Fault-Integrity Gating, and 30,000-Trial Monte Carlo Certification”

CRediT Role	Anoop V. Eluvathingal	K. Shanti Swarup
Conceptualisation	Originated the three-hypothesis SPRT framework for 87T; identified A_k as the separating statistic across inrush, IBR fault, and CT saturation	Provided the strategic research framing and guided novelty positioning relative to 2H-restraint literature
Methodology	Derived the truncated-Normal distributions for $H_0/H_1/H_2$ from first principles (Jiles–Atherton, IBR controller bandwidth, CT knee-point); designed the Wald boundary parameters; formulated the f_{int} index and proved the 0.34 pu formal separation (Proposition 1)	Reviewed all derivations; confirmed alignment with IEC 60255-111 and IEC 60255-187 standard requirements
Software	Implemented the full SPRT algorithm and five-priority relay logic in Python/ANDES; coded the 30 000-trial MC engine across four groups; developed TR-57a/b and TR-64 test harnesses	—
Validation	Ran all 48 deterministic scenarios and all MC trials; performed Wilson CI analysis; verified zero-overlap of f_{int} distributions across 5 000 inrush trials	Independently verified the KL-divergence computations and the formal separation proof
Formal Analysis	Proved Proposition 1 (0.34 pu trip/restrain separation); derived the expected decision time formula from Wald’s approximation	Reviewed the proof for mathematical rigour and physical interpretation
Investigation	Designed three-stage simulation trilogy (TR-57a/b/c + TR-64); identified the four algorithm refinements (R1–R4); characterised the SG DC-transient worst case (220 ms)	Advised on transformer CT class selection and magnetisation parameters
Writing — Original Draft	Wrote the full manuscript	—
Writing — Review & Editing	—	Reviewed and edited all sections; approved the final submission
Supervision	—	Supervised the PhD programme of which this paper forms a part
Funding Acquisition	—	Secured SERB and IIT Madras research funding

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Anoop V. Eluvathingal: Conceptualisation (lead) — identified waveform asymmetry index as the separating statistic across all three 87T hypotheses; Methodology (lead) — derived all truncated-Normal distributions from physical models, designed Wald boundaries, formulated and proved the 0.34 pu formal separation; Software (sole); Validation (lead) — ran 30 000 MC trials and 48 deterministic scenarios; Formal Analysis (lead) — proved Proposition 1 and expected decision time formula; Investigation (lead) — designed three-stage simulation trilogy and identified R1–R4 algorithm refinements; Writing — original draft (sole).

K. Shanti Swarup: Conceptualisation (supporting) — strategic research framing; Methodology (supporting) — reviewed all derivations and confirmed standard compliance; Validation (supporting) — independently verified KL divergence computations and separation proof; Formal Analysis (supporting) — reviewed Proposition 1; Writing — review and editing (sole); Supervision; Funding acquisition.